

COURSE CONTENT

UNIT I: Introduction

Survey of industrial catalytic processes, Theories of heterogeneous catalysis, Classification of catalysts and supports

UNIT II: Catalyst preparation techniques

Impregnation, adsorption, sol – gel, chemical vapour deposition, Factors governing catalytic activity.

UNIT III: Catalyst characterization

Bulk characterization, Physio-sorption technique, Surface area and pore size distribution, Chemisorption, Desorption spectroscopy, Thermal analysis, X-ray diffraction, Electron microscopy, Electron spectroscopy

UNIT IV:

Synthesis, characterization and modification of micro-porous and meso-porous zeolites, Hetero-poly compounds, Pillared clays

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UNIT V: Deactivation

Mechanism and kinetics of activation, deactivation and regeneration of catalysts; Sintering of supported metal catalysts

UNIT VI: Electro catalysis

Electro catalysis and fuel cell, Photo-catalysis for the removal of air and water pollutants and conversion of solar energy; Poly-functional catalysts

UNIT VII: Nanoparticles

Synthesis, characterization and activity of nano-particles and nano-clusters

Unit VIII: Other topics

TEXTBOOKS:

1. Thomas, J.M. and Thomas W.J., Introduction to Principles of Heterogeneous Catalysis”, Academic Press, (1967)
2. Butt, J.B. and Peterson, E.E., “Activation, Deactivation and Poisoning of Catalysts”, Academic Press, NY(1988)
3. Anderson, R.B. (Ed.), “Experimental Methods in Catalytic Research” Volume I to III, Academic Press, (1968)